

Awotoro Ebenezer Oladimeji

Researcher & Software Engineer

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CORE ENGINEERING

Python PHP Laravel Lumen Django
Flask JavaScript Vue.js Node.js

DATA & RESEARCH

Machine Learning Deep Learning
Bioinformatics AMR Surveillance Proteomics
Data Visualization

SYSTEMS & DELIVERY

REST APIs Microservices Docker Git
CI/CD Scrum TDD SOLID

DATABASES

MySQL PostgreSQL MongoDB
Oracle Database

LANGUAGES

English Fluent
German Conversational
Yoruba Native

RESEARCH METHODS

Unsupervised Learning Tensor Decomposition
Network Analysis Survival Analysis
RNA-Seq Analysis AMR Surveillance

KEY TOOLS

PyTorch scikit-learn Pandas Jupyter
Structurizr Postman Linux

PROFILE

Researcher and software engineer, PhD candidate at the Robert Koch Institute / Freie Universität Berlin, with 6+ years delivering production backend systems and 3 years specialising in ML-driven public-health surveillance. Previously Technical Lead at SeamlessHR, leading multi-module platform delivery for Nigeria's leading enterprise HR product. Proven across roles that bridge rigorous backend engineering, high-dimensional data research, and cross-functional technical leadership.

EXPERIENCE

Doctoral Researcher Oct 2024 – Present

Robert Koch Institute · Berlin, Germany

- Building ML and visualization systems for Germany's national AMR surveillance programme at the Robert Koch Institute.
- Analyse multi-year, multi-pathogen surveillance datasets to detect resistance trends and co-detection patterns across national data.
- Develop AI-driven anomaly detection and visualization pipelines integrated into RKI's public-health reporting workflows.
- Apply tensor decomposition, unsupervised learning, and high-dimensional categorical methods to complex epidemiological data.

Research Assistant Apr 2023 – Sep 2024

Robert Koch Institute · Berlin, Germany

- Designed and implemented ML pipelines for large-scale epidemiological datasets spanning multiple pathogen types and resistance classes.
- Evaluated classification and clustering algorithms for AMR phenotype prediction; contributed methods carried forward into doctoral research.
- Teaching assistant at Freie Universität Berlin; supervised and assessed graduate students in data science and bioinformatics courses.

Back End Engineer, Technical Lead Nov 2021 – Mar 2023

SeamlessHR · Lagos, Nigeria — Nigeria's leading enterprise HR SaaS platform

- Led cross-functional team of four engineers, UI/UX, QA, and product manager delivering HRIS and Leave modules used by enterprise clients.
- Authored C4 architecture models in Structurizr and technical specifications that aligned engineering, product, and QA across sprint cycles.
- Owned backend release readiness end-to-end: defect triage, code review standards, deployment coordination, and QA sign-off process.

Back End Developer Mar 2019 – Sep 2021

Casafina Development Company · Lagos, Nigeria

- Built and launched consumer loan and savings platforms from the ground up as lead backend engineer on a two-person team.
- Established TDD practices and CI/CD pipelines that improved release reliability and reduced regression incidents.
- Produced API and architecture documentation covering system design, onboarding, and client integration workflows.

Back End Developer, Contract Mar 2021 – Aug 2021

Patricia · Lagos, Nigeria — Concurrent freelance engagement

- Delivered WordPress/PHP customisations — themes, plugins, and third-party API integrations — for a fintech client within a 6-month engagement.
- Conducted cross-browser and cross-device QA to ensure compatibility before client handover.

Menaget Inc · Lagos, Nigeria — Concurrent freelance engagement

- Delivered Laravel backend services, REST APIs, and database schemas for a property management platform.
- Resolved performance bottlenecks and security vulnerabilities; introduced code review practices for the junior engineering team.

SELECTED PROJECTS

MetaMP Server

Python · Bioinformatics · REST API

Robert Koch Institute · Research Software

Backend research platform for membrane-protein metadata enrichment, AI-assisted analysis, and visualization workflows. Supports the MetaMP publication currently under review.

<https://github.com/Ebenco36/MetaMP-Server>

AMR Surveillance Workflows

AMR · Unsupervised Learning · High-Dimensional Data

Robert Koch Institute · Doctoral Research

ML and visualization pipelines for national antimicrobial resistance surveillance, covering co-detection pattern analysis, rare pathogen trends, and anomaly-oriented exploration.

SeamlessHR Platform Modules

Laravel · Vue.js · REST APIs · Technical Leadership

SeamlessHR · Enterprise HR SaaS

HRIS and Leave modules serving enterprise clients. Led backend delivery from technical specification through production, coordinating engineers, QA, and product.

<https://seamlesshr.com/>

Menaget Client Platforms

Laravel · PHP · REST APIs · Database Design

Menaget Inc · Contract Engineering

Built and maintained Laravel backend services, REST APIs, and database schemas for client-facing property management and e-commerce platforms.

<https://www.esarcar.com/en/>

PUBLICATIONS & RESEARCH OUTPUT

UNDER REVIEW

MetaMP: Seamless Metadata Enrichment and AI Application Framework for Enhanced Membrane Protein Visualization and Analysis

Ebenezer Awotero et al.

UNDER REVIEW

Network-based characterisation of latent co-testing patterns in antimicrobial susceptibility testing

Ebenezer Awotero et al.

IN PREPARATION

Cascade analysis of antimicrobial resistance patterns in surveillance data

Ebenezer Awotero et al.

EDUCATION

2024 – Present

PhD Candidate

Freie Universität Berlin / Robert Koch Institute, Germany

Doctoral research on ML, high-dimensional data analysis, and visualization for AMR surveillance.

2019 – 2022

Master of Science, Bioinformatics

University of Ibadan, Nigeria

Applied ML to ischemic stroke RNA-Seq data; identified differential gene expression patterns.

2012 – 2015

Bachelor of Computer Application

Bangalore University, India

Computer software technology and application development foundation.